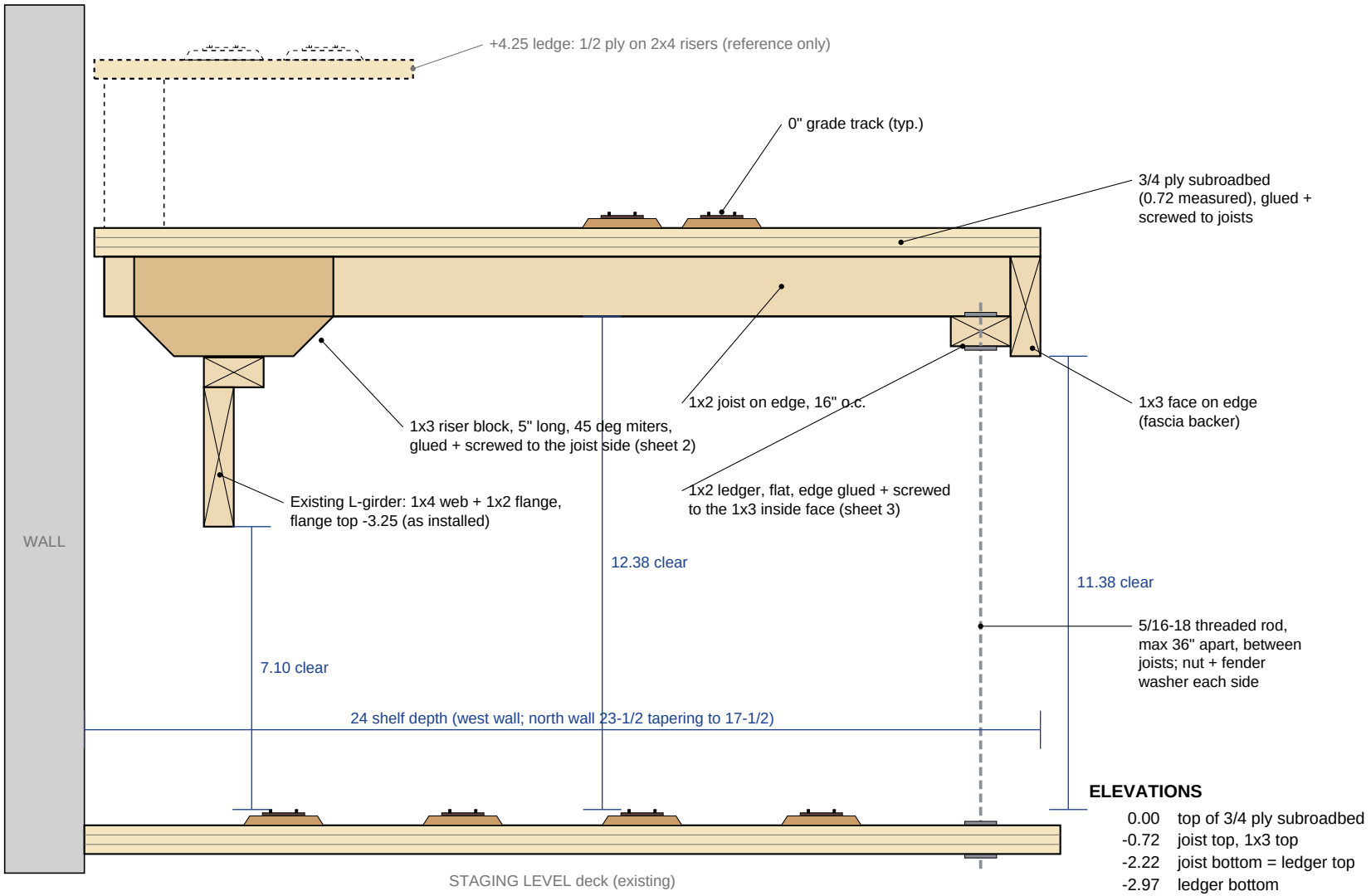
# Typical shelf section - north & west walls



Section through a joist, perpendicular to the wall.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).




**ELEVATIONS**

| | |
|---|---|
| 0.00 | top of 3/4 ply subroadbed |
| -0.72 | joist top, 1x3 top |
| -2.22 | joist bottom = ledger top |
| -2.97 | ledger bottom |
| -3.22 | 1x3 bottom, block bottom |
| -3.25 | girder flange top |
| -7.50 | girder web bottom |
| -14.60 | staging railhead |
| -15.00 | staging deck top |

Block bottom lands 0.03 above the flange (3.25 - 2.50 - 0.72); the deck top finishes 1/32 low - within tolerance, or shim the flange.
Girder drawn 3" off the wall; use the as-installed position. Track = 0.236 cork + 0.160 Atlas code 83 (caliper before clearance sign-off).
Scale 1:4 on Letter landscape. Lumber cut in section shows an X; the joist and riser block are in side view.

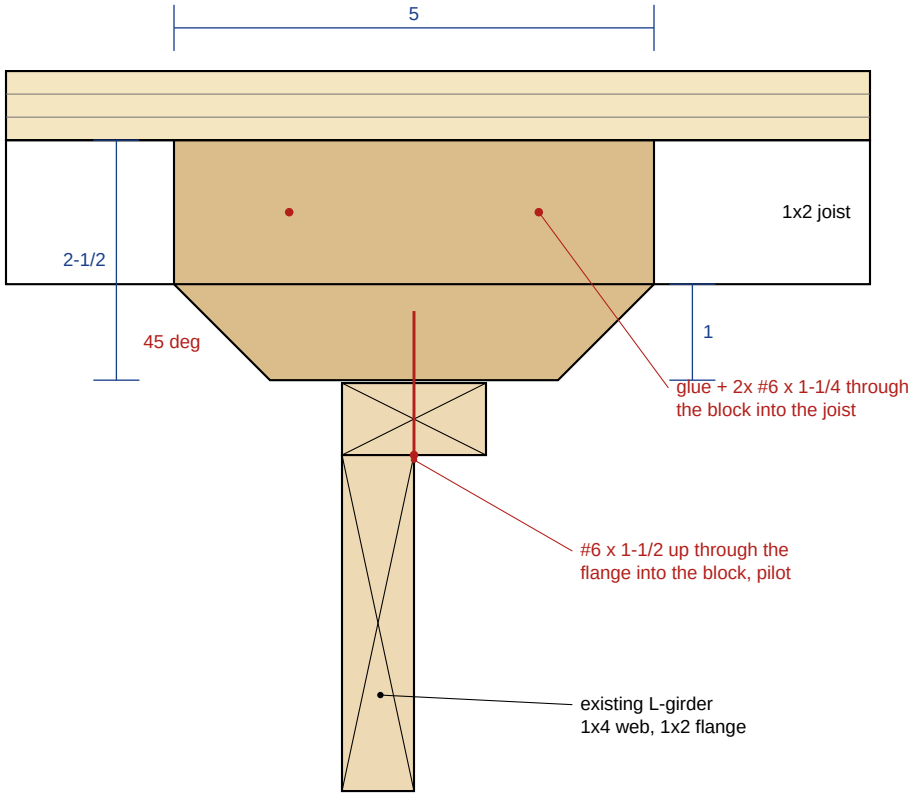
# Details - riser block at the back girder



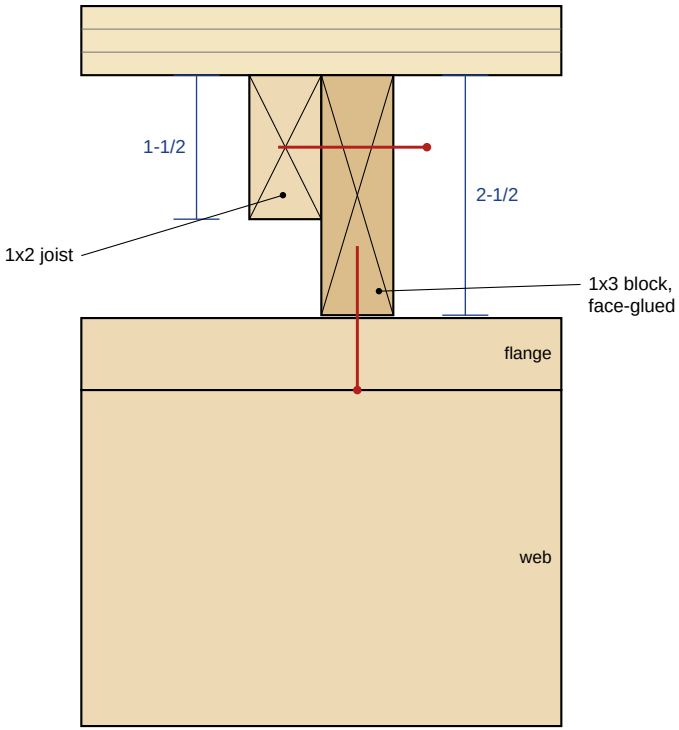
A: side view along the joist. B: looking along the joist (joist and block cut, girder in elevation).

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

### A SIDE VIEW ALONG THE JOIST 1:2




### B LOOKING ALONG THE JOIST 1:2




Riser block: cut from 1x3, grain running with the joist. Top flush with the joist top; the bottom sits on the flange. The 45 deg miters only trim the 1" hang-down, so the full 3" bottom still bears on the 1-1/2 flange. Keep every block on the same side of its joist. Full-size pattern on sheet 4.

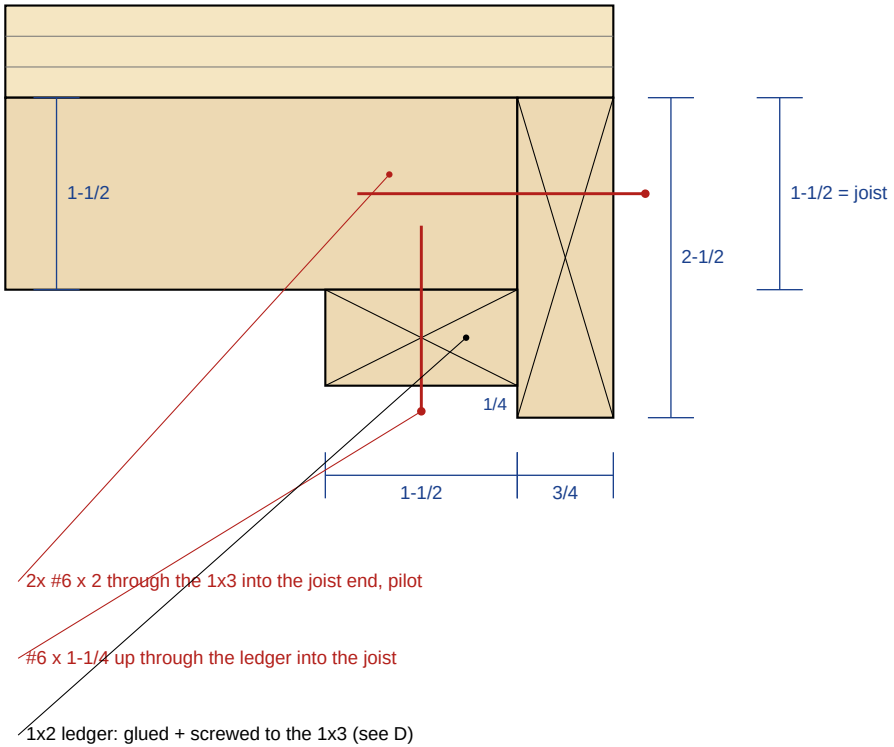
# Details - front 1x3 face + 1x2 ledger



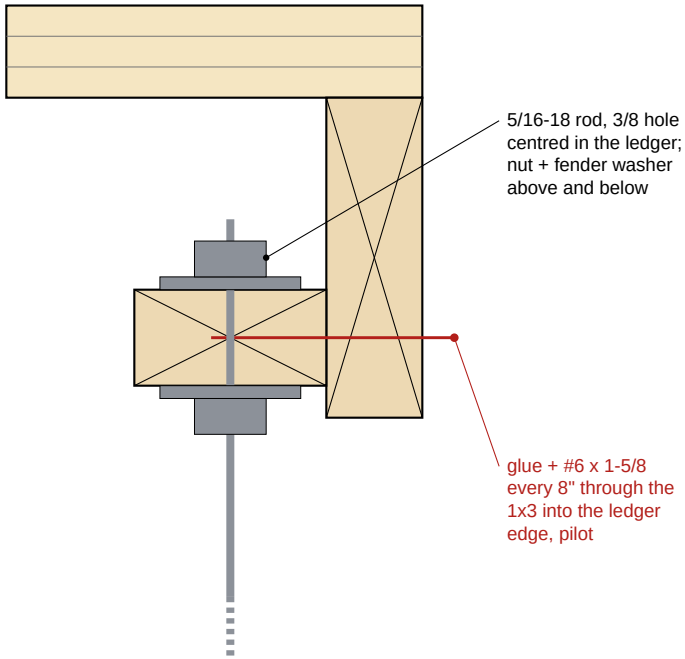
C: side view along a joist at the front edge. D: the same edge between joists, at a rod.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

## C AT A JOIST 1:1.5




## D BETWEEN JOISTS, AT A ROD 1:1.5



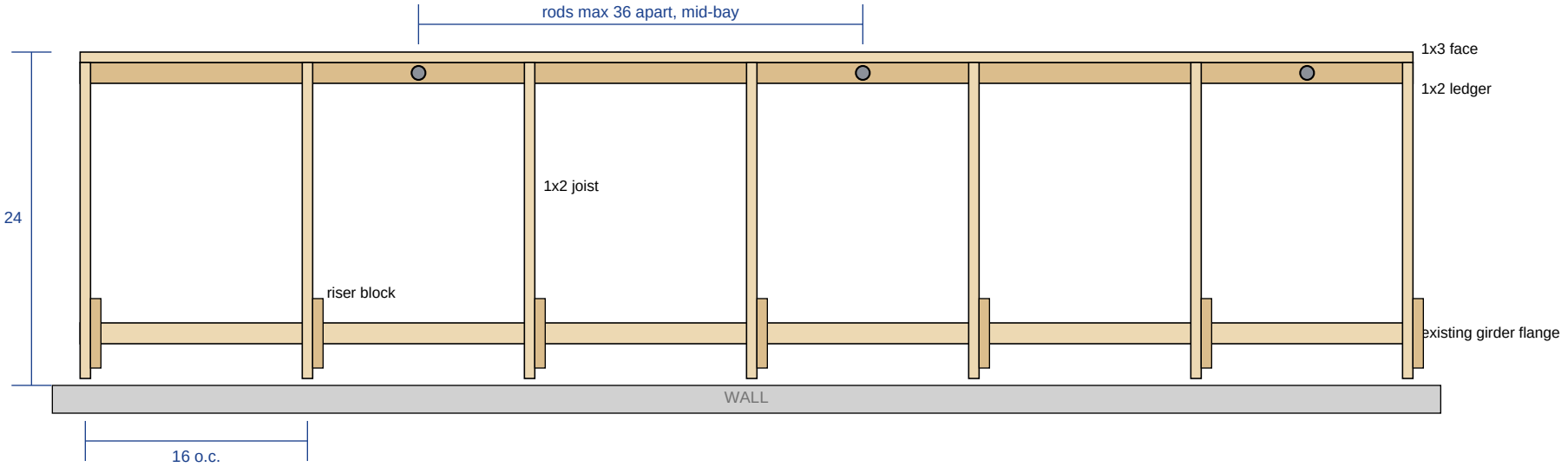

to staging deck (nut + washer each side)

Build the face + ledger on the bench first. Set the ledger height with a 1x2 offcut resting on it, its top flush with the 1x3 top: ledger top 1-1/2 below the 1x3 top, 1/4 up from its bottom. Ledger-to-face is a long-grain glue joint (edge to face); the glue is the structure and the screws clamp it while it cures. Stagger the face and ledger splices.

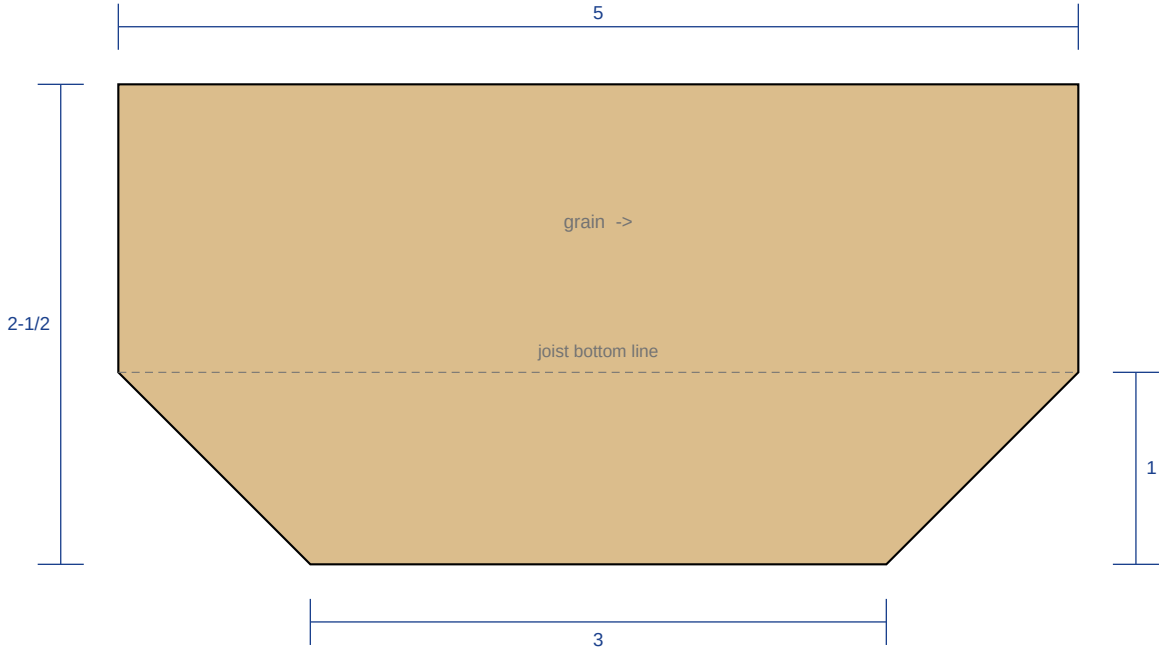
# Plan - typical 8 ft bay, deck removed + riser block pattern



Looking down. Joists at 16" o.c. land on the existing girder via the riser blocks; rods carry the front ledger between joists. Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).




## RISER BLOCK - FULL SIZE PATTERN (1:1)




## CUT LIST PER 8 FT BAY (24" deep)

| | | |
|---|---|---|
| 1x2 joists | 7 @ 22-3/4 | 24 - 3/4 face - 1/2 wall gap |
| 1x3 riser blocks | 7 @ 5 | 45 deg, 1" up each end |
| 1x3 face | 96 | on edge |
| 1x2 ledger | 96 | flat, edge to the 1x3 |
| 5/16-18 rod | 3 @ ~13 | fit to the staging deck |
| 3/4 ply | 24 x 96 | one rip per bay |

Splice the 1x3 face and the ledger at different joists; butt joints in the 3/4 ply land on a joist. North wall: 23-1/2 deep tapering to 17-1/2 - cut each joist to depth - 3/4 - 1/2.
Scales: plan 1:12, pattern 1:1 - print at 100% and check the 5" dimension with a rule before tracing.

# Assembly sequence and fastener schedule



Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

1. Rip the front members: 1x3 face and 1x2 ledger to length. Stagger splices so they never fall at the same joist.

2. On the bench, glue the ledger flat to the inside face of the 1x3, its edge against the face. Set the height with a 1x2 offcut on the ledger, flush with the 1x3 top (ledger top 1-1/2 below the 1x3 top, 1/4 up from its bottom). Screw from outside every 8". Let it cure.

3. Cut the riser blocks: 5" of 1x3, then trim each bottom corner at 45 deg, 1" each way. Trace the full-size pattern on sheet 4 or make a stop block on the miter saw.

4. Mark where each joist crosses the girder. Glue + screw a block to the side of each joist, top flush, centred on the flange line. Same side on every joist.

5. Prop the front assembly at height (ledger top at -2.22). Drill 3/8 holes in the ledger mid-bay for the rods, max 36" apart, between joists.

6. Set the joists at 16" o.c.: blocks on the flange, front ends on the ledger, butted to the 1x3. Screw up through the flange into each block, through the 1x3 into the joist end, and up through the ledger into the joist.

7. Run the rods from the staging deck to the ledger. Level the front edge with the nuts, then lock with the opposing nuts.

8. Dry-lay the 3/4 ply; butt joints on joists. Glue + screw every 8" along each joist, keeping screws out from under the track centrelines (feeders).

9. Check the deck top against the helix exit and the +4.25 ledge risers before the glue sets.

## FASTENERS

| | |
|---|---|
| Block to joist | glue + 2x #6 x 1-1/4 |
| Flange up into block | #6 x 1-1/2, pilot |
| Ledger to 1x3 face | glue + #6 x 1-5/8 @ 8" |
| 1x3 into joist end | 2x #6 x 2, pilot |
| Ledger up into joist | #6 x 1-1/4 |
| Ply to joists | glue + #6 x 1-1/4 @ 8" |
| Rod at ledger / deck | 5/16 nut + fender washer each side, 3/8 hole |

## CLEARANCE OVER STAGING

| | |
|---|---|
| Under joists | 12.38 in |
| Under front 1x3 | 11.38 in |
| Under girder web | 7.10 in |

Railhead = staging deck + 0.236 cork + 0.160 track.